

GOOGLE CYBERSECURITY CERTIFICATE

Linux File Permissions

Google Cybersecurity Professional Certificate · Study Guide

Auditing and correcting Linux file and directory permissions across a research team's projects directory — reading permission strings, and using `chmod` to fix over-broad access on regular files, a hidden archive file, and a directory.

Track	Google Cybersecurity Professional Certificate
Course	Assets, Threats, and Vulnerabilities
Environment	Linux Bash shell
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1 Reading Permission Strings

The research team's projects directory needed a permissions audit — existing permissions didn't match the required authorization levels. `ls -la` lists every entry including hidden files, with a 10-character permission string at the start of each line.

Position	Characters	Meaning
1st	d or -	d = directory, - = file
2nd–4th	rwx	User (owner): read, write, execute
5th–7th	rwx	Group: read, write, execute
8th–10th	rwx	Other: read, write, execute

EXAMPLE `project_t.txt` has `-rw-rw-r--`: a file, user and group can read/write, other can only read, and nobody has execute permission on it.

2 Correcting Permissions with chmod

Remove write access for "other"

Policy: no file should be writable by "other". `project_k.txt` had this issue.



o-w removes write from other

```
$ chmod o-w project_k.txt  
$ ls -la
```

Fix an archived hidden file

`project_x.txt` is archived — no one should retain write access, but user and group should keep read access.



u-w removes user write · g-w removes group write · g+r adds group read

```
$ chmod u-w,g-w,g+r .project_x.txt  
$ ls -la
```

Restrict a directory

Only `researcher2` should retain execute access to the `drafts` directory; the rest of the group should lose it.



g-x removes execute from group (researcher2's own access is unaffected)

```
$ chmod g-x drafts  
$ ls -la
```

3 Quick Reference



chmod — quick reference

```
ls -la                # list all files including hidden, with permission strings
chmod o-w <file>       # remove write from "other"
chmod u-w,g-w <file>   # remove write from user and group
chmod g+r <file>       # add read for group
chmod g-x <dir>        # remove execute from group on a directory
```